

DEPARTMENT OF MINING ENGINEERING

Mine Ventilation Practical-II

(MNC 303)

Mine Ventilation Laboratory



I.I.T (ISM), DHANBAD

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Experiment-1

Aim:Determination of susceptibility of coal to spontaneous combustion using Crossing Point Temperature method.

Apparatus Required:-

- Crossing point temperature apparatus
- Thermometer
- Source of O₂ supply

Theory:

Spontaneous Combustion:

Spontaneous combustion or spontaneous ignition is a type of combustion that occurs by self-heating (increase in temperature due to exothermic internal reactions) of coal or other carbonaceous material, followed by thermal runaway (self-heating which rapidly accelerates to high temperatures) and finally, ignition.

The occurrence of spontaneous combustion in coal mines may be seen when the following three conditions are available:

1. Coal must be in a form that can oxidize readily at ambient temperature.
2. Oxygen must be available to support oxidation.
3. Favorable conditions for accumulation of heat must be available.

Crossing Point Temperature method: Crossing point temperature (CPT) is the temperature at which the coal temperature coincides with that of the furnace temperature or bath temperature in °C. CPT or critical oxidation temperature gives an idea about the proneness of coal to auto-oxidation or self-heating. This is a standard method followed in India for finding out the susceptibility of coal to spontaneous combustion. The higher the crossing point temperature value, the less will be the susceptibility of coal to spontaneous heating.

Table 1: Risk rating criteria based on CPT values

CPT, °C	Spontaneous combustion susceptibility
120-140	Highly liable to self-heating
140-160	Moderately liable to self-heating
>160	Less liable to self-heating

This method envisages heating coal samples in an oxidizing atmosphere in which oxygen/air is passed at a definite rate through a bed of coal powder immersed in a glycerin bath heated to raise the temperature at a constant rate. Initially, the temperature of the coal bed shows a lower value than

theP a g e | 2 bath temperature. As the oxidation of coal is an exothermic reaction, it evolves heat and the temperature of the coal bed increases gradually. The oxidation rate is enhanced, and a time comes when the temperature of the coal bed equals that of bath temperature and then exceeds it. The point at which coal bed temperature equals the glycerin bath temperature is calledthe crossing point temperature or critical oxidation temperature of the coal concerned. CPT depends on whether the bath is heated up at a constant temperature or an increasing temperature. The increasing bath temperature is considered to represent better simulation as, in actual cases of spontaneous heating, the coal accumulates exothermic heat. Thus, the temperature of the surroundings increases gradually.

This is one of the oldest approaches for determining susceptibility toward spontaneous heating of coal. It is the temperature at which the coal temperature coincides with that of the furnace temperature or bath temperature in °C. In this method, the coal sample is heated in a furnace within a reaction tube at a constant rising temperature with oxygen passing through it at a predetermined rate till the coal temperature crosses the furnace temperature.



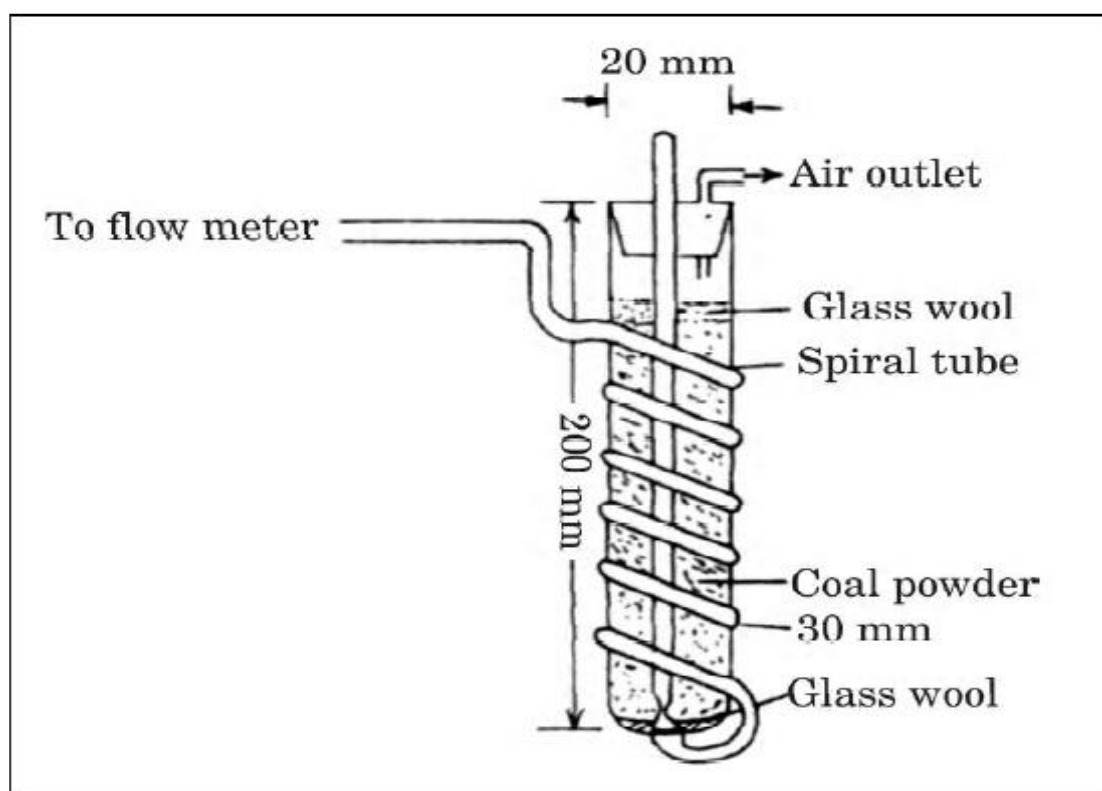
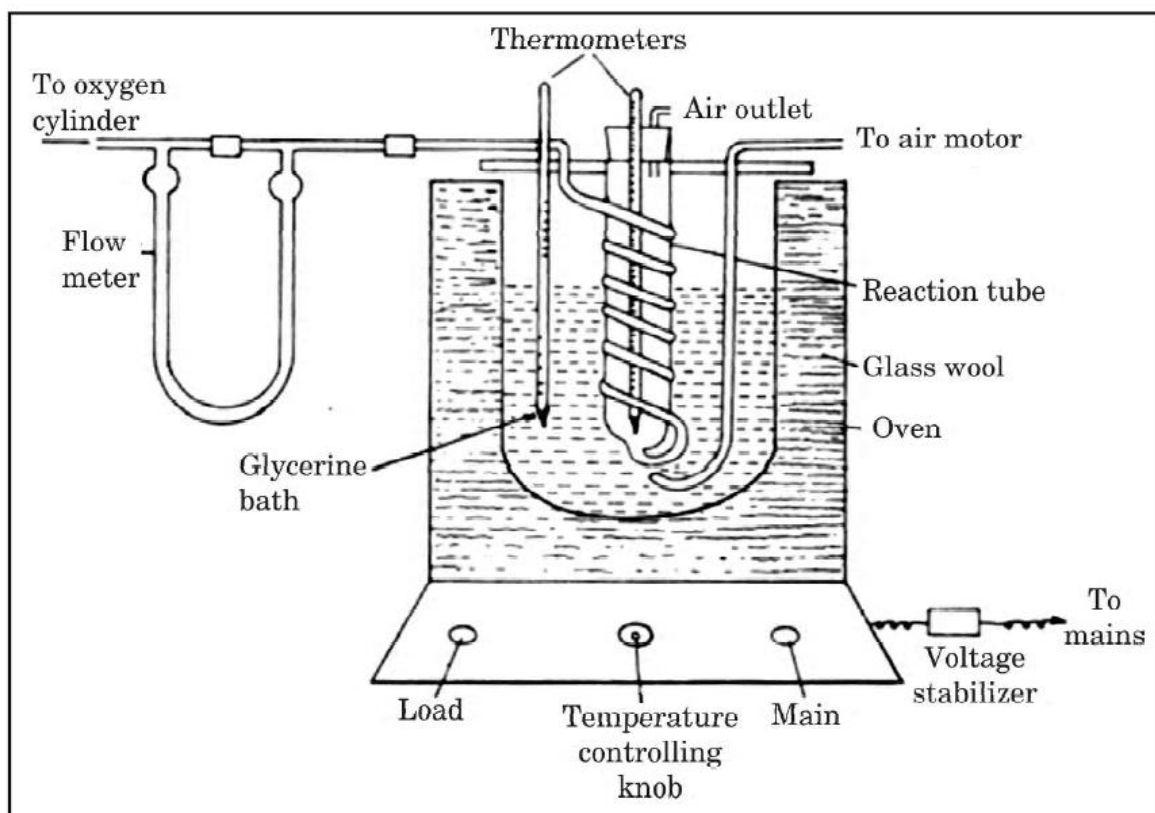


Figure: Photograph and schematic diagrams of the Crossing Point Temperature Apparatus

Procedure:

- 20 gm of coal sample of size -212 micron is placed in the reaction tube where glass wool is followed at the bottom most part of glycerin bath.
- The tube is then lightly tapped with the help of cork a fixed number of times to achieve uniform packing density of the samples.
- Then the reaction tube is placed in the tubular furnace, and a thermocouple is inserted at the center of the sample for the CPT experiment.
- After switching on, the furnace air is allowed to pass through the sample simultaneously with an average heating rate of 1 °C per minute, and airflow is 80ml/ min. The furnace and the coal sample temperatures are recorded at every five-minute interval until their temperatures are equal, as shown in the following figure.

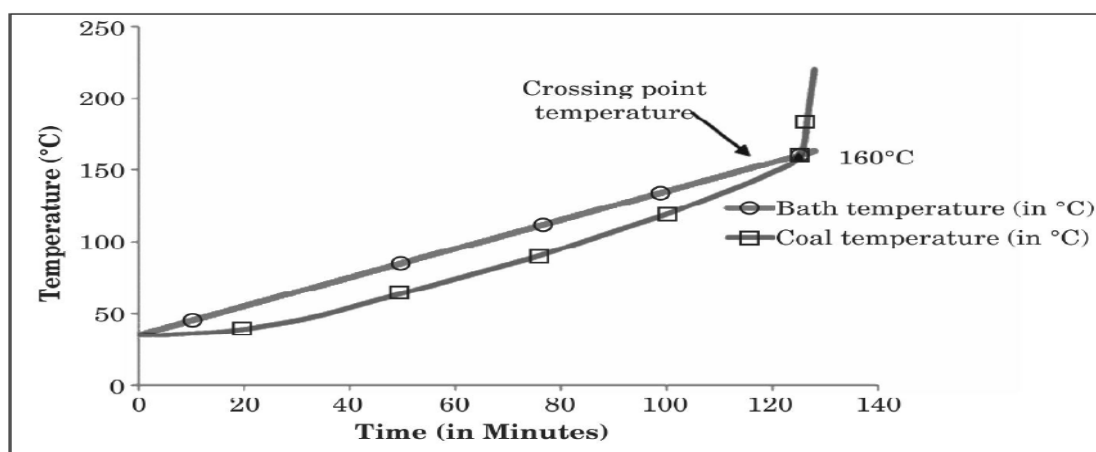


Figure: CPT for a coal sample

Observation table:

Time, min	Bath temperature, °C	Coal temperature, °C	Remarks
0			
5			
10			
15			

Result: The temperatures of the glycerin bath and the coal sample are noted at equal time intervals and plotted on the graph paper.

Limitations: The major drawback of CPT experiment is that the rate of flow of O₂, rate of heating, the particle size of coal, *etc.* have to be appropriately maintained. Otherwise, the CPT may change with slight change in any of the above conditions. This method is very time-consuming and takes 3-4 hrs. To complete the experiment and failure of the power supply even for a few minutes during experimentation, especially when the temperature approaches CPT, gives erroneous results and the whole experiment is to be repeated

Experiment-2

Aim: -Determination of Inflammability Index of a given coal sample by Godbert Greenwald apparatus.

Apparatus Required:

- GodbertGreenwald apparatus
- Oxygen Cylinder

Theory:

Flammability or explosibility index of coal

A unit weight of coal is dispersed in an environment of air or oxygen and ignited to cause coal dust explosion. When the amount of inert dust is mixed with the above coal dust, it renders it non-flammable and measures the explosibility of coal ssamples. The explosibility factor or simply the explosibility of a coal-dust deposit (E_x) is given by

$$E_x = \frac{100}{100 - S}$$

Where S is the amount of total inert matter (percent) that must be present in a unit mass of coal dust to prevent flame propagation through the mixture as determined by test.

Description of God Bert – Greenwald apparatus

The furnace comprises a heated silica tube capable of reaching temperatures of 1000 °C. Built-in control, trip, and sample thermocouples are used to measure the internal temperature of the Silica tube. For compactness, a sample powder injection module is close coupled to the furnace inlet and is mounted via the furnace sidewall assembly. The bottom of the furnace is open to the atmosphere allowing combustion products to vent into the fume cupboard. A stainless steel mirror is placed at a suitable angle and underneath the furnace to allow visual monitoring of the sample as it enters the heated Silica tube. The furnace, stand, and associated support assemblies are stainless steel, while the sample powder injection module is Brass.

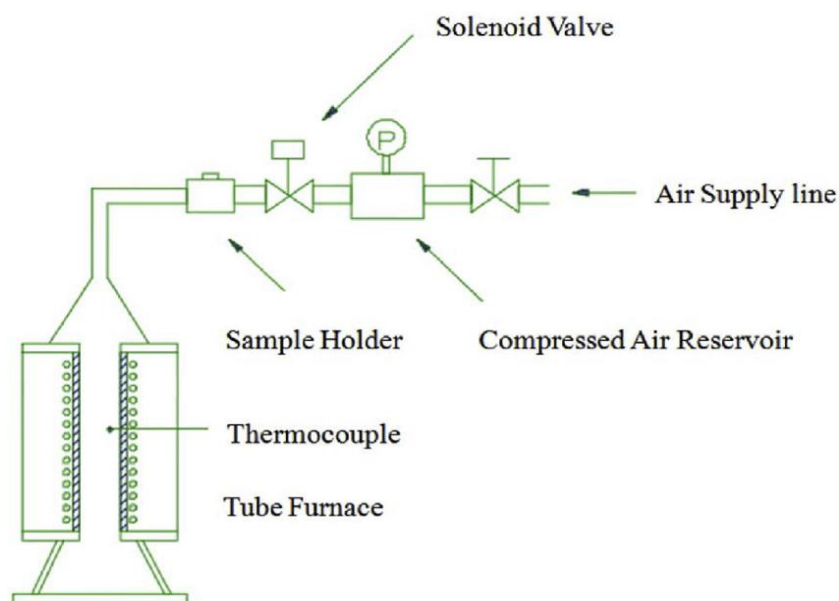


Figure: Photograph and schematic diagram of Godbert–Greenwald apparatus.

Air Input Assembly

A reservoir is pressurized via a regulated airline and then vented into the powder injection module to inject the sample powder into the heated tube assembly. This pulse of air carries the sample powder into the heated Silica tube. The air input assembly comprises an air reservoir complete with inlet/outlet

valves and a pressure gauge. This assembly can be assembled and customized for the installation as required.

Control / Monitoring Instrumentation

Control instrumentation for the god Bert-Greenwald furnace has been split into three modules, Temperature control / Air Reservoir Control, Heater Power Control, and Remote Control Trigger.

Temperature Control: This module is responsible for setting and controlling the Silica Tube's internal wall temperature. It also monitors over temperatures and automatically removes power from the heating elements of the Silica Tube if over temperatures are detected. The inside wall of the Silica Tube is also monitored. Hence, the surface temperature of the tube is known. Eurotherm controllers and indicators are used for all control and display functions.

Air Reservoir Control: Manual control over the air inlet and outlet valves to charge and discharge the air reservoir can be carried out via this instrument. A timer function has also been employed to control the reservoir discharge valve's open time, allowing the valve to fully open or partially open to discharge the air reservoir.

Power Control Module: The power control module comprises a main transformer and solid-state relay to control the Silica Tube heating elements. This module can be mounted locally to the furnace as required.

Remote Control Trigger Module: This module allows the operator to open the reservoir outlet valve while looking for an event inside the Silica Tube via the furnace mirror. A push-button located on the remote trigger module is used to open the air reservoir outlet valve. Addition features on the remote module allow the operator to manually charge the air reservoir and adopt a semi-automatic mode where the air reservoir is charged with air after each venting.

Procedure:-

- (i) Ensure that the control thermocouple is in thermal contact with the heater tube by observation only.
- (ii) Using the thermocouple extension leads connect the dual thermocouple in the furnace to the instrument's control input and sample input. Also, connect the single thermocouple to the trip input.
- (iii) Connect the transformer unit mains plug to the HEATER OUTPUT-protected mains outlet. Next, connect the yellow 110v mains furnace plug to the socket.
- (iv) Switch the controller mains POWER SWITCH ON.
- (v) Apply power to the oven by pressing the RESET pushbutton after the required temperature trip level on 3216 trip temperature is set.

-
- (vi) Once the furnace temperature has stabilized, switch on the air supply, set the pressure regulator to 5 psi, and adjust VALVE OPEN TIME to 0.25s. Next, check the operation of GG furnace remote control unit.
 - (vii) The fume cupboard interlock should be tested after installation and periodically after that.
 - (viii) The equipment is now ready for minimum ignition temperature (MIT) dust cloud testing.

Observation Table:

% of stone dust	% of coal dust	Whether flame observed or not	Explosibility factor

Results and conclusion:-

The explosibility factor is calculated based on the percentage of stone dust in the coal dust-stone dust mixture at which no flame was observed.

Experiment-3

Aim: Determination of airborne respirable dust concentration using Gravimetric Dust Sampler (Model-MRE113A) and Konimeter.

Apparatus required:-

- Gravimetric Dust sampler (Model-MRE 113A)
- Konimeter

Theory

MRE Gravimetric Dust Sampler:-

The gravimetric dust sampler is a self-powered portable instrument. It is used in the gravimetric (by the difference in weights) sampling of respirable airborne dust in mines. This is hung in a mine roadway at the height of 2/3rd of the full height of the roadway. The instrument operates for a full shift and collects harmful dust particles on a filter. From the difference in the weights of the filter, the harmful airborne dust per cubic meter of air is calculated.

A rechargeable 7.5 V, 90 mA battery drives the governed motor and a pump to draw 2.5 l/min of dust-laden air through a horizontal elutriator size selector. The harmful dusts are collected on the filter. The counter measures the amount of air sucked based on the number of liters of air inspired by the instrument during the sampling. The counter is resettable by the knurled accessible through the door of the instrument case. The pump is of diaphragm type and is fitted with two flap valves. A flowmeter has three horizontal lines engraved. The middle line represents the correct flow rate of 2.5 l/min, and the one below represents an underflow of 2.5 l/min, whereas the above line shows an overflow of 2.6 l/min. The filters used are 5.5 cm circles of the glass fiber or membrane type. The membrane filter is much more convenient to use.

The amount of airborne respirable dust (mg/m^3) =
$$\frac{1000(W - w)}{N - n}$$

Where w is the initial weight of the filter paper (mg), W is the final weight of the filter paper (mg), n is the initial reading of the counter, and N is the final reading of the counter. The amount of dust collected (mg) is $W - w$, and the air quantity sucked (liter) is $N - n$.

Description of the gravimetric dust sampler: The MRE 113A gravimetric dust sampler comprises basically a size selector, a filter holder and an electrically driven pump designed to pass 2.5 liters of air per minute. The coarse particles in the dust laden air settle down by gravity in the elutriator while the harmful fine particles in the respirable size range are collected on a filter of 55 mm in diameter. Glass fiber or membrane filters in a light aluminum holder are used for sampling. The amount of air in liters

sucked during sampling is recorded on a counter. The counter is resettable to zero by the knurled knob through the door of the instrument case. The pump, diaphragm type, is fitted with two flap valves and a flow smoother at the outlet. A flowmeter is connected to the output side of the pump. The counter and the flowmeter are readable through small windows fitted in the side of the instrument. The window of the instrument has three horizontal lines engraved, the middle line representing the correct flow rate of 2.5 l/min.

The size selector is a precision-made multi-channel horizontal elutriator mounted on the top of the instrument. A slotted restrictor plate is fitted at the mouth to minimize the effects of air cross-flow, and the rear of each selector plate is lipped to prevent accidental shaking onto the filter of the dust collected in the size selector. The transfer hood between the filter holder and the size selector is held in position by two screws. Removal of this hood gives access to both the filter holder and the size selector for cleaning and blowing out accumulated dust.

Two interchangeable horizontal elutriators are available. The first with four channels provides a performance such that there is a 50% collection of particles with a falling speed equal to that of a 5 μ unit density sphere. The second with eight channels provides a 50% collection of particles with a falling rate equivalent to a 3.5 μ unit density sphere.

The rear curved end of the instrument case is hinged and retained shut by a socket head screw, opening this end exposes the filter holder. The filter is fitted by unscrewing the knurled and winged ring and inserting the filter into the gap between the upper and lower components until it sits in the recess. It is then clamped by screwing up the knurled ring. This causes the two halves of the holder to come together and grip the filter amount firmly the 'O' rings.

A small cylinder lock is fitted to the case door, which exposes the battery, counter and fuse holder when opened. The battery is a DEAC 5/900D with a special internal limiting resistor added, and it is held in its compartment by a lockable sliding spring clip.

The counter is coupled by gearing to the pump unit and registers the number of liters of air inspired by the instrument. It is resettable by the knurled knob accessible through the door of the instrument case. The pump is diaphragm type and is fitted with aluminum-coated polyester film flap valves. An adjustable crank varies the pump output, and the unit is driven through super gears at half motor speed by a governed 6V 3-pole electric motor. A flowmeter is connected to the output side of the pump through a diaphragm type smoothing device. The counter and flowmeter are readable through small windows fitted to the side of the instrument case.

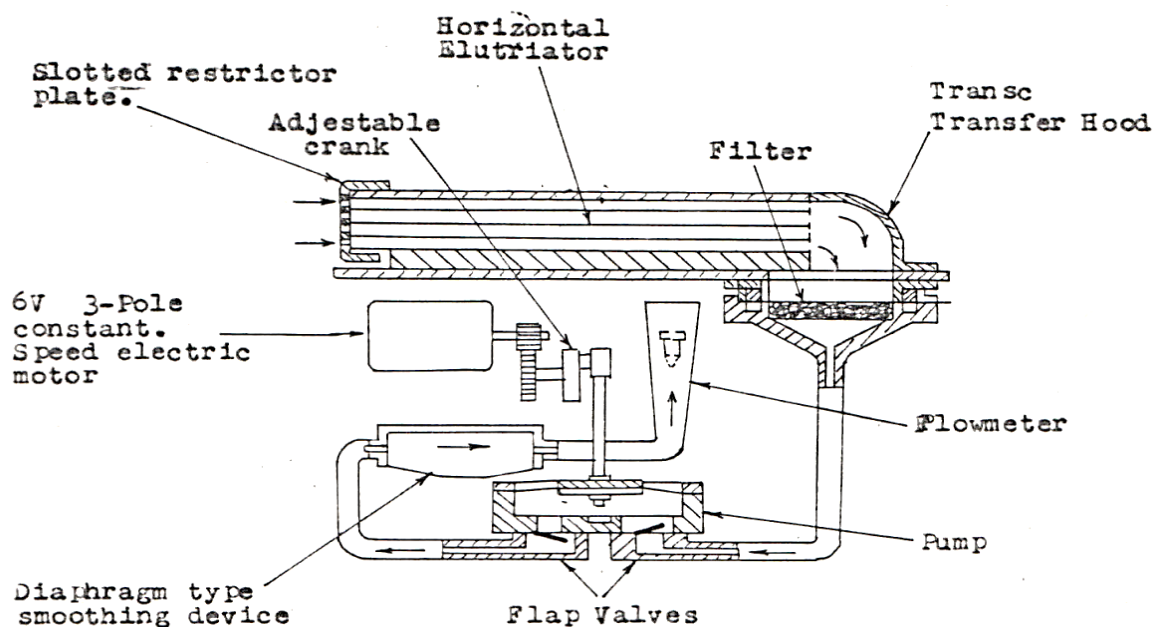


Fig. . Schematic diagram of M R E Gravimetric dust sampler.

Figure: Photograph and schematic diagram of Gravimetric Dust Sampler (Model: MRE 113A)

Procedure:

- A clean filter paper is selected and mounted in the filter holder. The assembly is weighed in an accurate balance having a sensitivity of 0.1 mg. It is inserted into the sampler and tightened in position. The battery is charged fully by the charger provided. A nearly discharged battery needs 14 hours charging. The charged battery is inserted into the instrument, and the count is set to zero.
- The gravimetric dust sampler is then placed where the breathable dust concentration is to be determined at a height approximately equal to the breathing level of the workers and switched on.

- Usually the duration for sampling covers a whole shift, but the time can be less for high dust concentrations or more for low dust concentrations so that sufficient amount of dust should be made at the sampling rate, which should be around 2.5 l/min. A lower flow rate of dust-laden air indicates overloading of the filter paper with dust or discharged battery.
- After sampling, the volume of air sampled is noted. The dust-laden filter paper is removed from the instrument and its folder and weighed. The difference in weight (in mg) divided by the volume of the air sampled in m³ gives the dust concentration in mg/m³.

Konimeter:

Description of Konimeter: The Konimeter consists of a spring-operated plunger pump, a dust deposition chamber, a glass plate, an air inlet jet, and a built-in microscope. Konimeter, however, can give a quick spot reading of dust concentration in PPCC (particles per cubic centimeter)

The components of the instrument are as follows:

- Pump:** The spring-operated pump sucks 5 cc of the air sample. Gun metal plunger fitted with two rings moves to and fro in a cylinder. The plunger locks itself automatically while at the bottom of the cylinder.
- Dust Deposition Chamber:** The lower half of the chamber carries a glass plate on which the dust is deposited. It is possible to rotate this in 30 stages around the center. On the upper portion, an air inlet jet is mounted through which the air sample is drawn into the dust chamber.
- Glass plate (50.8 mm × 4.34 mm):** The glass plate is a circular disc of heat resisting glass engraved with 30 sector markings. The plate rests at a fixed distance of 0.50 mm from the inlet valve and is made airtight by a rubber washer.
- Air Inlet Jet:** The air inlet jet consists of a carburetor-type jet (0.5 mm dia) with a parallel side passing through the upper portion of the dust chamber. This is protected by an outer rubber-lined metal cap which is lifted only when the sample is to be drawn in.
- Microscope:** This is a simple biological type with a total magnification of ×216 (approx.). For a rapid evaluation of the dust sample, a special graticule is fitted to the eyepiece on which are engraved intersecting lines forming two 18° sectors together with two parallel lines 5 microns apart.

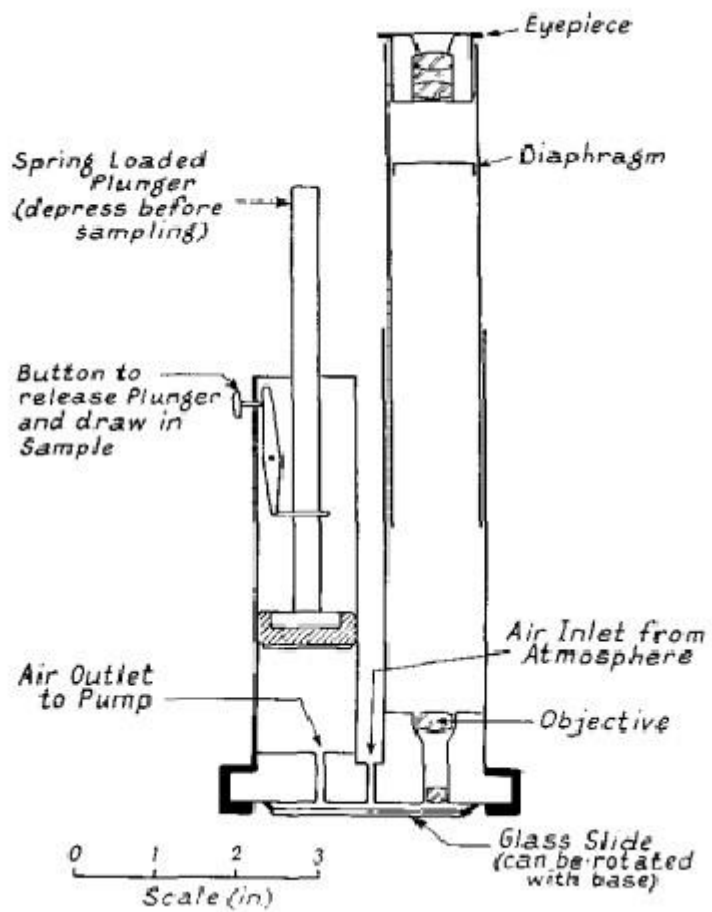


Figure: Photograph and schematic diagram of Konimeter

Procedure

1. Clean the glass plate with alcohol or some other organic solvents.
2. Treat the non-engraved side of the glass plate with a thin layer of petroleum jelly evenly.
Replace the glass plate in the proper position.
3. Depress the plunger until it is locked at the end of downstroke.
4. Hold the Konimeter at the desired place and lift the air inlet cover.
5. Turn the knurled ring slightly to release the plate for the automatic upstroke.
6. Collect the sample, if desired, by rotating the plate.
7. The total dust count can be calculated by

- Total dust count in 5 cc of the air sample = $\frac{360 \times N}{36}$
- Total dust count in 1 cc of the air sample = $\frac{360 \times N}{36 \times 5} = 2N$

Limitations & Accuracy

- a) The performance or efficiency under varying conditions is unpredictable. Different types of Konimeter give different results.
- b) It is difficult to evaluate a heavy deposit with high dust concentration.
- c) The quantity of dust deposited depends on the nature and thickness of the adhesive film.
- d) It tends to break up aggregates due to high impact velocity.
- e) Only spot sampling is possible.

Observation tables:

For Gravimetric Dust Sampler

Sl.No	Initial weight of filter paper (w), mg	Final weight of filter paper (W), mg	Initial reading of flowmeter (n), liter	Final reading of flowmeter (N), liter	Dust concentration (mg/m ³)
1					
2					

For Konimeter

Samples	Dust count of less than 5µm size		Total
	N ₁	N ₂	N ₁ + N ₂
1			
2			

Results and conclusions:

- Using gravimetric dust sampler, the dust concentration in terms of mg/m^3 can be calculated based on the initial and final weight of the filter paper, initial and final readings of the flow meter.
- Using Konimeter, the total dust count in terms of p.p.c.c. can be calculated by counting the number of dust particles deposited on the glass plate.

Experiment-4

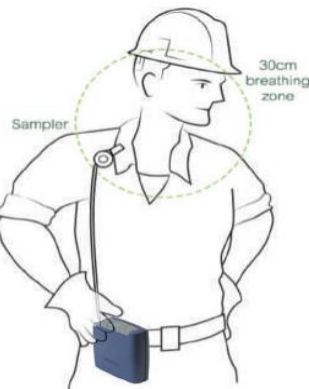
Aim: Determination of airborne respirable dust concentration using Personal Dust Sampler (Apex 2)

Apparatus Required : Personal Dust Sampler and Calibrator.

Principle: Personal Dust Sampler is a compact, self-contained device for collecting dust samples to measure workers' personal exposures to particular pollutants.

PDS Set Up as per Gravimetric Method

Instrument shall be provided as per below items to collect “Resperable Dust Fraction”



PDS Representation (How to fix)



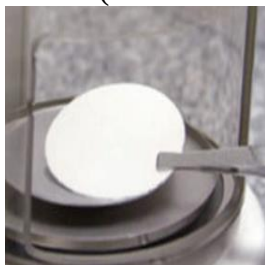
(Part No – APEX2IS Plus (Main Unit) & 209152B (Docking Station))



Higgins Dewell Cyclone (Part No - 116000B)



Filter Cassette (Part No – P104096)

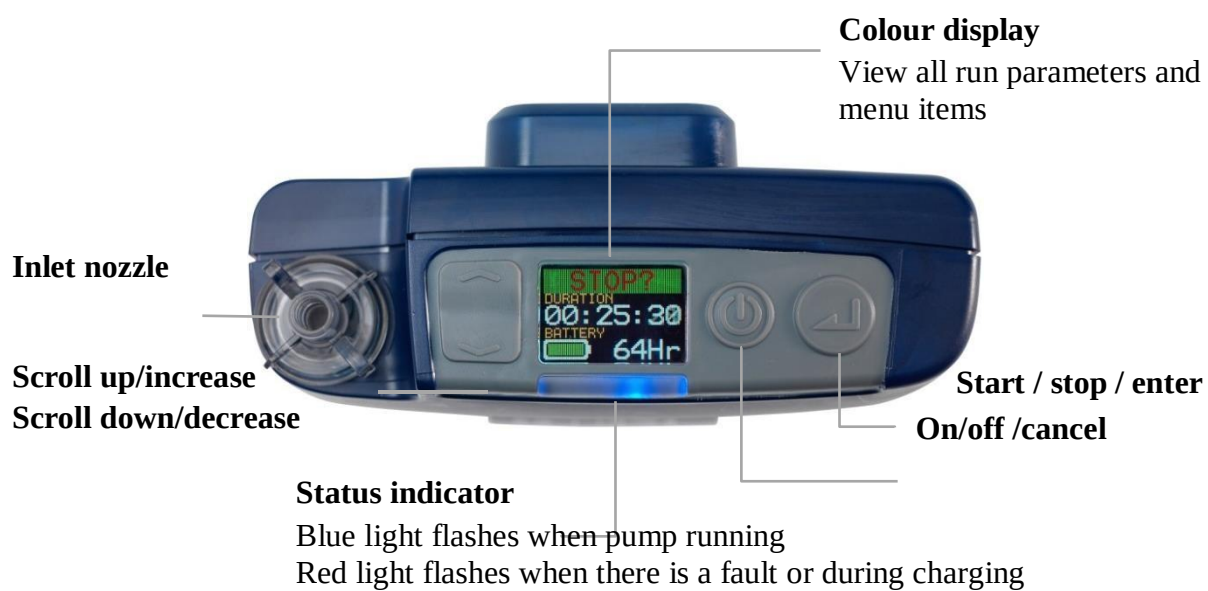


37mm PVC Filter Paper (Part No – P102079)



PVC ¼” 1 M Pipe for Dust/Air Sampling (Part No – TU104)

Construction:



Turn the pump on/off

To turn on the pump:

On the top of the pump, press . The sequence on the right is displayed:



This is followed by the three alternating dashboard screens, showing data from the last run. Notice that the screen header is prompting you to start a sampling run.



On the top of the pump, press and hold  until the countdown has finished and the screen goes blank.

Note: You can't turn off the pump during a sampling run or when a programme timer is set.

Setup before starting a run

Depending on your sampling requirements you may want to perform one or more of the following setup procedures before starting a sampling run.

To access the menu items described below:

1. Ensure the pump is turned on, and then press  to display the menu.
2. Press  or  to highlight the menu item you want to alter, and then press .

Set the flow rate

1. Navigate to **SETFLOW**, and then press .



2. Press  or  to change the flow rate, and then press the enter key.

Re set the data

Use this menu item to reset the accumulated run time, motion index and volume data to zero.

1. Navigateto **RESET**.
2. Press  for 3 seconds.



Calibrate the pump

Always calibrate the pump at the sampling flow rate for the greatest accuracy.

Note: If using a filter with a small pore size or if operating at a high flow rate, which results in a high back pressure, it may take the pump a little longer to stabilise when adjusting the calibrated value.

1. Attach a sampling head or flow tube or other flow-measuring device to the pump's inlet nozzle.
2. Navigate to **CALIBRATE**.
3. Press  to start the test.
4. Press  or  to change the pump Speed until the meter flow rate and the pump flow rate match.
5. Press  to stop the test.



Start/stop a sampling run

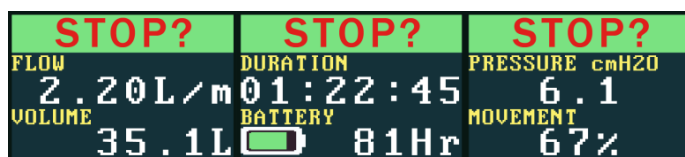
Ensure your pump is calibrated, set to the correct flow rate and you have reset the accumulated data if necessary.

To start a sampling run:

1. Turn on the pump.
2. Press  for 3 seconds.



The following screens should be visible.



To stop or pause a sampling run:

Press  for 3 seconds.



Lock and unlock the pump

Partial lock

Partial Lock mode is denoted by a half-open padlock icon and can be set in **Run** or **Stop** modes. While partially locked, the wearer may stop and start the sampling pump but has no access to the menu or other functions.



To partially lock and unlock the pump:

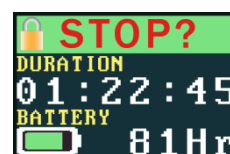
Press and hold , and then press  3 times in quick succession.

The partial lock icon will be displayed.

Note: If you try to remove a partial lock when the pump is running you will need to apply the above procedure twice, which applies and then removes a full lock.

Full lock

The full lock can only be set when the pump is running and is denoted by a closed padlock icon. While fully locked, the keypad is fully de-activated. The wearer cannot stop or disturb the pump by any key presses.



To fully lock the pump:

Press  and hold and then press  6 times in quick succession. The full lock icon will be displayed.

To unlock the pump:

Press  and hold and then press  3 times in Quick succession

Calculation

The average concentrations in mg/m³ is calculated using the following formula:

$$C = \frac{P}{V} \times 1000$$

Where P is the weight of sampled respirable dust (mg) and V is the total volume of filtered air (m³)

Experiment-5

Aim: Determination of airborne respirable dust concentration using Real-time Aerosol Monitor (Model: RAM 1) and Portable Aerosol Spectrometer.

Apparatus Required: Real-time Aerosol Monitor (Model: RAM 1) and Portable Aerosol Spectrometer.

Theory:

The RAM-1 is a portable, self-contained aerosol dust monitor whose sensing principle is based on detecting near forward scattered electromagnetic radiation in the near-infrared. The instrument uses a pulsed gas as a semiconductor light-emitting diode, generating a narrow-band emission centered on 940 nm. The scattered radiation is detected using a silicon photo-voltaic-type diode with an integral low noise preamplifier.

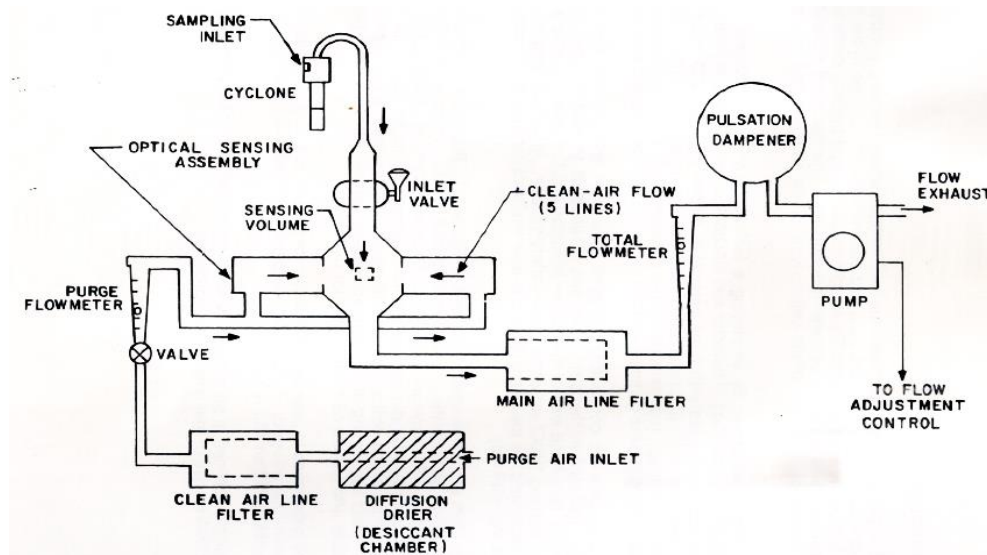
Signal processing is performed by employing a synchronous lock-in scheme which allows for continuous cancellation of detector and electronic circuitry drift and noise. The standard instrument is supplied with three selectable ranges (0-2, 20, 200 mg/m³) to allow for high resolution over a concentrated range of five decades. Further read-out flexibility is obtained through four selectable time constants (0.5, 2.8, and 32 s). Thus, the operator has complete control over both the range and speed of response.

The concentration data are continually displayed by a three and one half digit liquid crystal displayed. The display includes two diagnostics to alert the operator; a flashing "K" on the right hand side of the display indicates that the reference scattered is inserted, and a flashing "VDC" indicates a low battery voltage. An analog voltage (0-10 VDC) output proportional to the concentration is also available for strip chart recording, data logging and /or telemetry.

The flow system employs a diaphragm pump to produce 2 liters/minute sampling flow. A secondary clean air stream of 0.2 liters/minute continuously flushes filtered air over all critical optical surfaces. Also, during a zero check, the entire inlet airflow is filtered to effect a self-cleaning of the optical sensing chamber. The flowmeter are provided to allow for continuous monitoring of the total and filtered flows. All filter cartridges are readily accessible and require no disassembly for replacement.

The instrument is designed to operate continuously for about 6 to 8 hours after its internal battery has been fully charged. A separate charger is provided with which the RAM-1 can operate indefinitely from the AC line.





Photographs and schematic diagram of Real-time Aerosol Monitor
(Model: RAM 1)

Procedure :

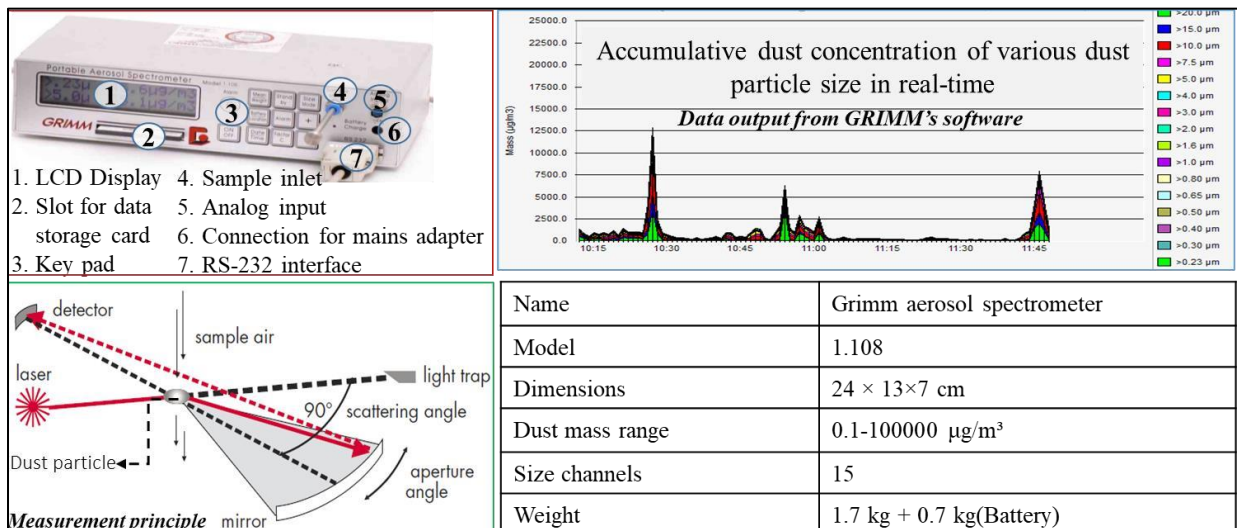
1. Switch on the Instrument (on /off switch at the top of the instrument)
2. Select concentration measurement range.
3. Select desired measurement time constant.
4. Place the inlet valve in the sample position (vertically downward)
(Note :- It is recommended that instrument should be always started in CLEAR position for about 01 minute to ensure proper purging prior to sampling)
5. Observe indication of the total flowmeter. It should indicate about 2.2 liters/minute
6. Observe indication of the Purge flowmeter. It should indicate approximately one tenth of the total flow rate, e.g. 0.22 liters/minute. The total flow rate minus the purge flow rate equals the inlet sampling flow rate.
7. The total flow rate should be adjusted by means of the flow screw adjustment (at the top of the instrument) while the purge flow rate should be adjusted by means of the small valve at the lower end of the purge flowmeter.
8. When the inlet valve is closed (Clear position), all the air flows through the clean – air purge line, and the indicator ball of the purge flowmeter will be pegged against its upper stop.
9. Dust concentration readings can now be obtained by viewing the digital display at the top of the instrument.
10. If the dust concentration exceeds the maximum selected range, the RAM – 1 will indicate 1 with all zeros blanked out.
11. During use, it is advisable to close the protective panel cover in order to protect the controls and display.

Measurement of airborne respirable dust concentration:

The real-time Aerosol monitor (Model: RAM 1) directly measures the respirable dust concentration on real-time basis spot reading in following ranges:

1. 0 to 2 mg/m³
2. 2 to 20 mg/m³
3. 20 to 200 mg/m³.

Portable Aerosol Spectrometer



2.2 Measuring principle

The sample air is led directly into the measuring cell via the aerosol inlet or other custom-designed air inlets, e.g. for high wind speeds or overpressure. The particles in the sample air are being detected by light scattering inside the measuring cell. The scattering light pulse of every single particle is being counted and the intensity of its scattering light signal classified to a certain particle size. The measuring principle is schematically shown in figure 1.

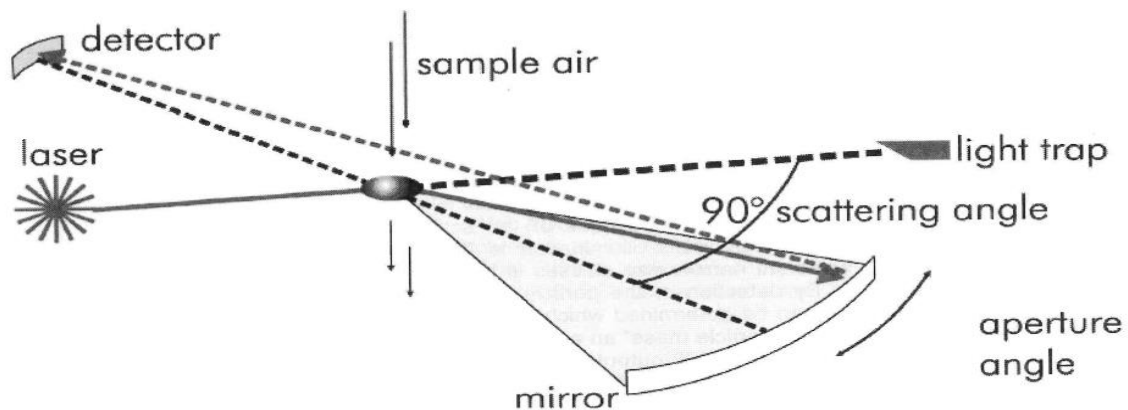


Figure 1: Measuring principle series 1.108 and 1.109

All Grimm laser aerosol spectrometers and dust monitors use a laser diode as light source. The wavelength is different within the model series and is for model 1.108 in infrared range at 780 nm and for model 1.109 in visible range at 655 nm. The laser diode can be operated in a so-called Multiplex Mode, i.e. the intensity of the laser beam is being modulated. This way particles can be detected over a very wide size range from 0.3 µm up to 20 µm (model 1.108) respectively 0.25 µm up to 32 µm (model 1.109). The laser beam is focused to a flat elliptical strip by means of illumination optics. Inside the focus the laser beam lights a small measuring volume evenly and subsequently is being led into a light trap. The sample air is focused aerodynamically and then led as particle flow through the inner area of the measuring volume. When doing environmental measurements, the particle concentration of the sample air is normally so low, that statistically seen only one particle is in the measuring volume. Measuring at particle sources, technical particulate matters, or working places, very high particle concentrations can appear which require a previous dilution of the sample air. Due to the fact that the entire sampling volume of 1.2 liter/minute is analyzed, all Grimm aerosol spectrometers reach a very good counting statistic. The scattering light emitted by every particle is being detected by a second optics under a scattering angle of 90° and then directed onto a receiver diode via a wide-angle mirror. The signal of the detector will be classified into size channels after amplification subject to its intensity. Figure 2 shows the assembly of the laser-measuring chamber. The sample air duct occurs perpendicular to the perspective into the measuring volume.

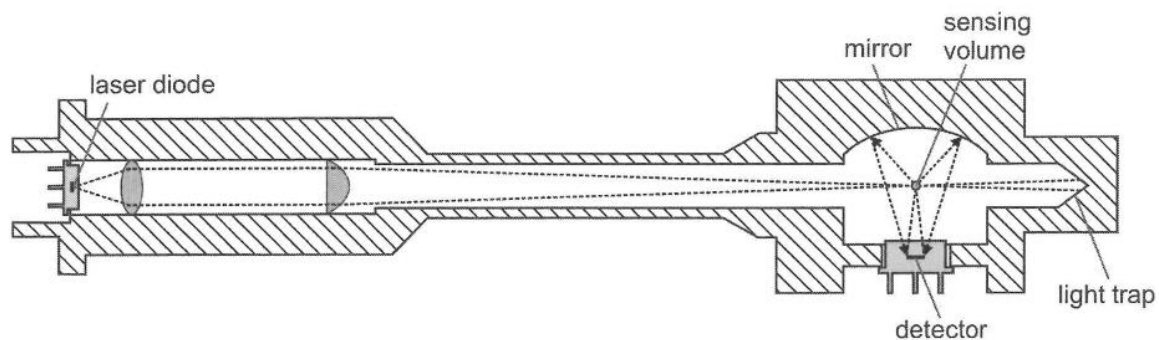


Figure 2: Laser measuring chamber series 1.108 and 1.109

The count rate results from the number of particles divided by volume flow rate. The particle size is proportional to the intensity of the detected scattering light signal whereas the scattering light intensity is also being influenced by the particles refractive index, particle shape, and orientation of the particle within the measuring volume. Positioning the detector into a 90° direction enables minimizing the influence of the aerosol particles' refractive index for determining the particle size. The opening angle of the detector optics was chosen in a way so that an ambiguousness of the scattering light intensity due to MIE scattering undulations caused by monochromatic illumination is being compensated. Thus a distinct detection of the particle size in sufficient narrow size classes is possible. Model 1.109 provides 31 size channels, model 1.108 offers 15. By detection of the particle concentration and particle size, the size distribution of the aerosol particles can be determined which in turn is the basis of the evaluation of the particle mass. Within measuring mode "particle mass" an additional size channel is arithmetically adjoined below the smallest size channel. So model 1.108 outputs within mass mode 16 size channels and model 1.109 analogical 32 size channels. The additional size channel is primarily useful when calculating the standardized mass fractions and leads to an improved accuracy measuring fine aerosols.

For a precise and reproducible particle counting and particle sizing the accurate sampling volume of 1.2 liter/minute is absolutely fundamental. The sampling volume affects the conversion of the raw counts in the actual sampling volume in the selected unit, like liter or m^3 and secondly the particles speed in the optical chamber, which must be in the given tolerance range for a correct particle sizing. That's why the sampling volume at the Grimm Aerosol Spectrometer Model 1.108 is controlled continuously.

Key functions in the standby mode

The settings of the aerosol spectrometer can be changed in the standby mode. by use of the [+] and [-] keys.

Mean values and calculated filter weight



The options for pressing the **[Mean/Weight]** key solitary are the same in the operational mode. Pressing the **[Mean/Weight]** key will display the mean value in mass or counts. The selected channels are hereby underlined. The mean calculation takes all measurements into account since the instrument was powered up or the mean value calculation was reset to zero.



When pressing **[Mean/Weight]** and holding the **[-]** key for 5 sec, the mean value will be reset to zero. Also the sample air volume will be reset to zero.



When pressing **[Mean/Weight]** and holding the **[+]** key for 5 sec the calculated filter weight and the corresponding sample air volume will be displayed. This value is essential for the determination of the C-factor for gravimetric validation.

Clear memory card and formatting the data storage card



When pressing **[Mean/Weight]** and the **[Factor C]** key simultaneously, the message "clear memory card, are you sure?" will show up. Confirming this question by pressing the key **[+]** will clear all data on the data storage card and format the card. If no data storage card is in the instrument, the message "clear memory, are you sure?" will show up. The erasure is the same procedure and will clear the 80 Kbytes internal memory.

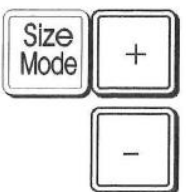
This key combination also lets you format a new data storage card previous to first time operation or if you want to use a data storage card which has been used in another device before.

Starting a new measurement



Leave the stand-by mode and start a new measurement by pressing the **[Stand-by]** key.

Setting the measurement mode and choice of the displayed size channels



Holding the key **[Size Mode]** and pressing additionally the key **[+]** or **[-]** lets you change the measurement mode, either particle mass or particle concentration. The LCD display shows on the right side the accordant unit $\mu\text{g}/\text{m}^3$ or particle per liter as "/l".

Decisive for the measurement mode and the change of the measurement mode is the setting in the upper line of the LCD display. Leaving the size range by pressing the key **[+]** or **[-]** the measurement mode changes.

Example 1: The display of the 1.108 shows in the mass mode **>20µmµg/m³** in the upper line. Pressing **[+]** again and the measurement mode will change to particle concentration, indicated by **>0.3µm /l**.

Example 2: The display of the 1.109 shows **>0.25µm..... /l** in the concentration mode, pressing **[-]** will switch to mass mode, indicated by **>32µm..... µg/m³**. The size channel that is chosen in the lower line of the LCD has no influence on the choice of the measurement mode. It simply shows the data of a second channel within operational mode.



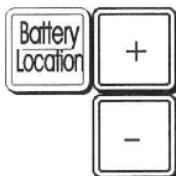
Please notice that the data sent via the RS-232 interface and the data stored onto the storage card are the same type as the set displayed operation type. That means either dust mass or particle concentration can be displayed and stored!

Displaying the analog values



Pressing only the key **[Size/Mode]** longer than three seconds or pressing additionally the key **[Date/Time]**, the values of the analog input values and time will be displayed. This remains set until pressing any other key.

Requesting the battery capacity and measurement location number

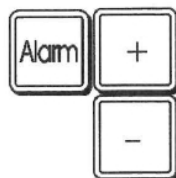


By pressing the **[Battery/Location]** key the current battery capacity as percentage as well as the set measurement location number will be displayed. When operating with a mains adapter, always 130% are displayed. At a fresh charged battery values above 100% are possible. Pressing and holding the key **[Battery/Location]** the measurement location number can be altered by means of the keys **[+]** or **[-]**. The measurement location number is an additional differentiation of various measurements and can be set between 1 and 99.



When reading out the data storage card via the Windows software, the datasets can be chosen individually according to their measurement location number, which shortens and simplifies the data transmission.

Changing the alarm thresholds, necessity of a diluter at coincidence

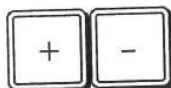


Simultaneous pressing of the keys **[Alarm]** and **[+]** or **[-]** changes the alarm thresholds. A value of 0 deactivates the alarm function. A value bigger than 0 activates the alarm. The alarm value always refers to the size channel in the upper line of the LCD and the set operational mode (mass or particle concentration).



An alarm value of 2 000 000 /l for the lowermost size channel ($>0,3\mu\text{m}$ for 1.108, respectively $>0,25\mu\text{m}$ for 1.109) shows within operation mode particle concentration a transgression of the measurable total concentration (coincidence). In this case a diluter is necessary.

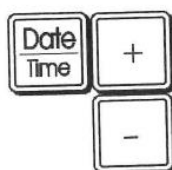
Setting the storage interval



storage time.

After inserting the data storage card the LCD shows the storage card's capacity as well as the spare storage time. During this display the storage interval can be changed with the keys **[+]** or **[-]**. By short lifting of the storage card and reinserting, the LCD again shows the storage card's capacity, but now with the new information of the spare

Setting date and time

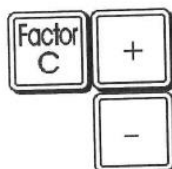


Short pressing: Displaying date and time.

Longer pressing: Setting date and time.

The location of the blinking cursor can be altered by pressing **[+]** or **[-]**. Pressing **[Date/Time]** again transfers the cursor. Pressing no key for about 3 seconds, the device switches back into stand-by mode. Changing the minutes zeroes the seconds.

Changing the gravimetric factor



Simultaneous pressing of the keys **[Factor C]** and **[+]** or **[-]** lets you alter the gravimetric factor in intervals of 0.05 between 0.1 and 9.9.



The C-factor will be taken into consideration when displaying the dust mass on the LCD. Displaying the dust mass via HyperTerminal they are always related to a C-factor of 1. The C-factor will be separately given on seventh position in the P-line (GF for gravimetric factor). When operating via the Windows software the C-factor has to be set in the menu "Options" under "Dust Monitor" respectively "Measurement Locations".

Printout of the storage card data



Within stand-by mode data can be transmitted to an external printer via the RS-232 interface by pressing **[Mean/Weight]** and **[Alarm]** at the same time. Only datasets that are concordant with the present measurement location number will be sent.

Key functions in the operational mode



In the operational mode, that is to say during active measurement, all settings are fix and can only be read by pressing the respective functional keys.

Mean values and calculated filter weight



The options for pressing the **[Mean/Weight]** key solitary, are the same in the standby mode. Pressing the Mean/Weight key will display the mean value in mass or counts. The selected channels hereby are underlined. The mean calculation takes into account all measurements since the instrument was powered up or the mean value calculation was reset to zero.



Pressing **[Mean/Weight]** and holding the **[-]** key for 5 sec will reset the mean value (mass or counts and sample volume), so the next mean value of the following measurement will be calculated without taking the previous ones into consideration.



Same as in the standby mode. By pressing **[Mean/Weight]** and holding the **[+]** key for 5 sec, the calculated weight on the filter and the corresponding sampled volume will be displayed. The estimated dust weight indicates whether the collected dust mass is sufficient to be determined within a suitable tolerance with the available microbalance.

The dust weight calculation takes all measurements into account (mass and counts) since the last filter change. It must be pointed out that the C-factor is not taken into account for calculating the estimated dust weight.

Stop a measurement



Pressing this key interrupts the measurement and switches the instrument to the standby mode. Pressing this key again re-activates the measurement process.

Hold the actual value on the LCD-Display



Typically the values are refreshed all 6 sec. and shown on the display. The actual value is holding on the LCD-display as long as the **[Size/Mode]** key is pressed while the measurement continues in the background. The operational mode can be changed in standby only.

Indicate the battery capacity and location number



Like in standby mode. By pressing the **[Battery/Location]** key the current battery capacity as well as the set measurement location number will be displayed.

Requesting the alarm thresholds



By pressing this key the current alarm value can be requested. The alarm value always refers to the size channel, which is set in the upper line of the display. Setting the alarm value to 0 this function is not active.



Pressing this key when using the weather protection housing shows the threshold of the relative humidity when the mixer operation shall be activated.

Behind these two concentric circles the speaker for the acoustic alarm signals is located.

Requesting the analog input values



Pressing this key **[+]** switches on the cyclic display of the analog input values of the hooked up sensors. With the key **[-]** the cyclic display can be switched off. The sensor values appear in the upper line of the LCD alternating with the measurands of the selected operational mode.

Requesting date and time



Pressing this key shows date and time.

Requesting the gravimetric factor



The set gravimetric factor will be shown when pressing this key. This also suppresses the cyclic display of the analog input values.

Alarm and error messages

A warning signal resounds and a message appears on the LCD display when:

- Exceeding the set alarm threshold
- Device errors (e.g. too low battery capacity, high pump motor current, etc.)

The errors will also be stored with the data sets onto the storage card and output via the RS-232.

Determination of the gravimetric factor (C-factor)

Please execute following steps:

1. Open the filter chamber and take out the old filter. Now clean the filter chamber e.g. with a Q-Tip. Also clean the sample air duct as described in chapter 6.3_Cleaning the optical chamber.
2. Weigh a new filter (at least three times) and note down the gained average weight.
3. Put the new, weighed filter centred onto the big O-ring of the filter chamber and close the chamber.
4. Turn on the dust monitor and respond to the question "Filter changed?" with YES [+] so that the calculated filter weight will be reset.
5. Execute your dust measurements at a characteristic location. Operate and transport the dust monitor if possible always with the LCD display facing upward.



Please mind, that in case of an eventual interruption of the measurement row the question for a new filter has always to be answered in the negative [-] when switching on the device.

6. The current calculated filter weight can be requested within operational or stand-by mode by pressing the keys [Mean/Weight] and [+]. This is also possible by a control command via HyperTerminal.
7. Having a filter weight heavy enough (at least 1 mg total dust mass) you can exit the measurement by switching into the stand-by mode and subsequently switching off the device.
8. Avoid agitations during transport.
9. Try to take out the filter without losing any collected material. Weigh the loaded filter for a higher accuracy again for at least three times. The difference to the empty filter weight is the effective gained dust mass.
10. Now you can calculate the gravimetric factor according to the equation below:

$$C - factor = \frac{\text{gravimetric dust weight on filter (loaded filter - empty filter)}}{\text{calculated dust weight (shown by instrument)}}$$

For improving the measurement accuracy one should do the C-factor determination several times. This value should not deviate from the basic value more than ca. $\pm 30\%$. However, at metalliferous dusts higher values can occur.



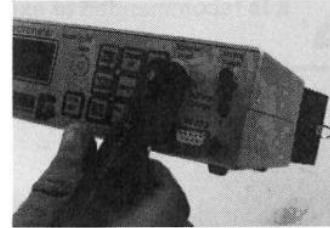
The calculated factor C can now be input into the dust monitor. All dust mass concentration values as well as their averages will be multiplied by this factor and accordingly corrected be output on the LCD display. The measurands on the storage card and the data output via the RS-232 interface will not be corrected. However the gravimetric factor appears in the headline. Thus the evaluation software arranges the necessary mass corrections automatically.

6.3 Cleaning the optical chamber

Turn off the device.

Open the filter chamber and remove the gravimetric filter.

First inspect the air passage optically for free sight through the instrument. The aerosol duct of the optical chamber is straight and thus you have to be able to see through it. Then flush the instrument with clean and oil-free compressed air (max. 3 bar). This must be done in direction of the airflow, only, from the air inlet located on the front side of the device to the filter chamber. Eventual blockage of the aerosol nozzle in the optical chamber is to be removed with a wooden or plastic rod (do not use any stronger materials in order to avoid damage of the nozzle!).



Dust Measurements are four Operational Mode

1. Environmental → PM 1 , PM 2.5 and PM 10
2. Occupational Health → Inhalable, Thoracic and Alveolic
3. Mass Distribution → 16 Channels ($> 0.23 \mu\text{m}$ - $> 20 \mu\text{m}$)
4. Count Distribution → 8 Channels ($> 0.3 \mu\text{m}$ - $> 2 \mu\text{m}$)
→ 8 Channels ($> 2 \mu\text{m}$ - $> 20 \mu\text{m}$)

Experiment-6

Aim: Study and sketch of self-rescuers (MSA IW-65 and other models)

Apparatus required: Self-rescuers (MSA IW-65 and Chemical Type)

Theory:

Self-Rescuer: The self-rescuer is a small emergency gas respirator designed to provide instant respiratory protection against CO gas in Mines that offer an explosion or fire and thus enable the wearer to escape to a place safely. It is compact and light in weight and is carried by the miner on his body. It should be used for entrance into an irrespirable atmosphere only when it is necessary to pass through it to get to an uncontaminated area. Self-rescuers do not offer protection in the atmosphere containing less than 16% oxygen. They can be used for 60min or more in concentration likely to be encountered after a fire or an explosion.

Filter Type Self-rescuer, MSA IW-65

Description: The IW65 Self-Rescuer Respirator provides emergency respiratory protection against carbon monoxide gas produced from underground mine fires or explosions. The respiratory should always be carried so that it is ready for immediate use. The Self-Rescuer is a one-time device and must be discarded after use. Its weight is low, nearly 1 kg.

The respirator should be used immediately at the first sign of a fire or explosion-even if no smoke is visible. If the air is oxygen-deficient (less than 19.5%), or there are other toxic gases and vapors present, the Self-Rescuer may not provide adequate protection. In such situations, a self-contained breathing apparatus must be used.

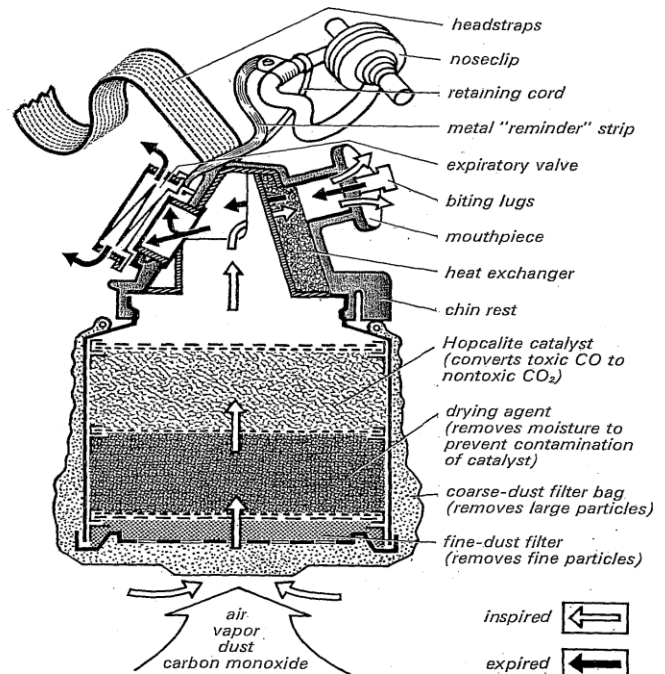
Principle of Operation:

The self-rescuer's filter section contains Hopcalite, a mixture of manganese dioxide and copper oxide, an oxidation catalyst. This material changes toxic carbon monoxide into non-toxic carbon dioxide. The filter section also has an outer coarse-dust filter, an inner fine-dust filter to remove dust particles, and a drying agent to protect the Hopcalite from moisture. The filter materials are separated by screens and baffles. The drying agent is charcoal impregnated with a mixture of calcium bromide and lithium chloride in the proportion of 10:1 (approx. 53% CaBr₂ and 5% LiCl).

Inspired air is cooled by an integral heat exchanger before inhalation. Expired air passes back through the heat exchanger and out through the spring-loaded expiratory valve. Excess saliva is also expelled through the expiratory valve. The inhalation resistance at 95 l/min is approximately 6 mbar, while the exhalation resistance is about 1.8 m bar.



Principle of operation



Photograph and schematic diagram of filter type self-rescuer (MSA IW65)

Drawbacks: The IW65 Self-Rescuer is an air-purifying device designed to protect the wearer from toxic carbon monoxide. It does not supply oxygen but functions to convert carbon monoxide to carbon dioxide with a resulting heat of reaction. Therefore, when the Self-Rescuer is worn in an atmosphere containing carbon monoxide, the air entering the wearer's mouth will be hot and dry. The inhaled air temperature depends upon the concentration of carbon monoxide – the higher the concentration, the higher the temperature. The very presence of heat indicates that carbon monoxide is present in the air drawn into the Self-Rescuer. Although the person using the self-rescuer feels uncomfortable due to hot and dry air, the Self-Rescuer should never be discarded. All air must be drawn through the Self-Rescuer.

The IW65 self-rescuer contains a heat exchanger to reduce the discomfort caused by high concentrations of carbon monoxide. For example, at 1.5% carbon monoxides, the heat exchanger would effectively reduce inhaled air temperature from approximately 149 °C to 65 °C. Though uncomfortable, one can tolerate an even higher inhaled air temperature since the respiratory system is an effective heat exchanger.

Chemical type self-generating oxygen self-rescuer

Introduction:

RakshaKavach is a single used self-contained closed-circuit system oxygen-type breathing apparatus. RakshaKavach will assist a person in escaping from an irrespirable atmosphere containing smoke, toxic gases, or an oxygen-deficient atmosphere. As it is a closed-circuit apparatus, its operations do not depend on the surrounding atmosphere.

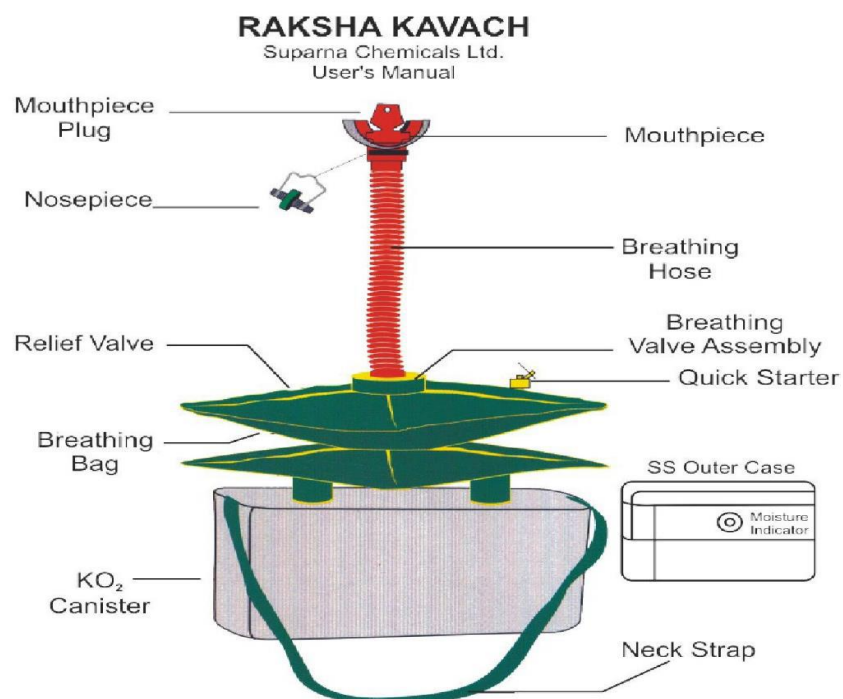


Figure: Photograph and schematic diagram of chemical oxygen type self-rescuer (RakshaKavach)

(2) Applications and Limitations:

It is a lifesaving apparatus. It is designed to escape any difficult and life-threatening condition, such as an environment of toxic gases, smoke & or oxygen-deficient atmosphere. It is used in emergency in:-

- (a) Underground mines, particularly in coal Mines.
- (b) Sub Marines
- (c) Many branches of industry.

RakshaKavach is designed for single use. It has got DGMS approval for 30min. duration.

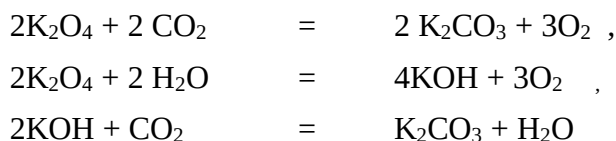
(3) **Design:**

Components:-

- (i) One stainless steel canister containing KO₂.
- (ii) One 6 liters breathing bag with relief valve.
- (iii) One breathing valve assembly.
- (iv) One breathing hose with nose piece, mouth piece and mouth piece plug.
- (v) One neck strap.
- (vi) One heat protection &
- (vii) Suitable heat exchanger.

(4) **Principle of Operation:-**

The unit is activated with one or two breaths of the user. He inhales CO₂ and water vapors into the KO₂ bed of the apparatus through the breathing hose. KO₂ reacts with water vapors to produce oxygen and simultaneously remove CO₂ from the inhaled gas. The wearer gets oxygen free of CO₂. This chemical reaction is demand sensitive. The more frequently the user breaths, the more water vapor and CO₂ gases inside the KO₂ bed & more oxygen is generated. At rest, breathing slows down, and therefore oxygen generation is also less, and KO₂ bed life is also more.



(5) **Operating Instructions:-**

- a) Remove unit from the belt.
- b) Lift the latch on the top.
- c) Remove bottom & top covers.
- d) Remove helmet.
- e) Put nose piece to close both nostrils.
- f) Remove mouthpiece plug from mouthpiece.
- g) Inhale ambient air through the mouth.
- h) Inhale into the mouthpiece.

- i) Repeat the process 2 to 3 times to fill the breathing bag.
- j) Insert & hold mouthpiece in bag.
- k) Breathe normally.
- l) Put on safety goggles.
- m) Adjust neck strips and hang breathing apparatus comfortably.
- n) Put on helmet.
- o) Move out.

(6) **Precautions:-**

- a) RakshaKavach will be used by persons who are trained to use this equipment.
- b) The unit is a completely sealed unit and it is for single use only. Never open the latch of the unit out of curiosity. Its effectiveness may go down.
- c) Use only for escape in emergency.
- d) While using, do not remove mouth piece and nose dip unless you come to a safe place.

Experiment-7

Aim: Study and sketch of self-contained oxygen breathing apparatus (Model: BG-174A and BG-4).

ApparaatusRequired : BG 174, and BG 4

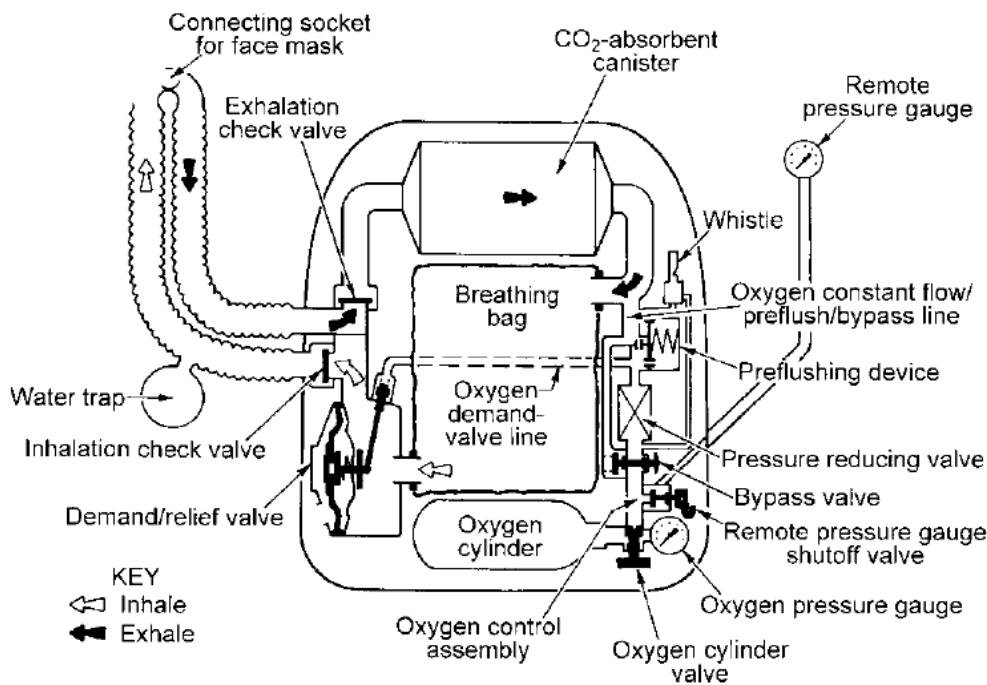
Dragger BG-174A

- It is a compressed O₂ type closed-circuit apparatus with a constant and governed dosage of O₂.
- Manufactured by Dragger AGCO, Lubbock, Germany.

Main Specifications:-

- Safe working period: 4 hr. hard work.
- Cylinder capacity (empty): 2 liter.
- Cylinder capacity with O₂ at 200kg / cm²: 400 lit.
- Breathing bag capacity : 6 lt. (approx.)
- Weight of apparatus (fully charged): 12.8 kg.
- O₂ flow rate: 1.5 lit/min.
- O₂ feed by lung governed value: As req. by users.





Photograph and schematic diagram of Dragger BG 174A

Special features of the apparatus BG-174A:

- The O₂ regulator comprises a pressure reducer, an automatic pre-flushing device, a bypass valve, and a pressure gauge shut-off valve.
- The lung-demand valve is a diaphragm-operated stem valve in a valve box or housing and operated by the diaphragm responding to small negative pressures regulating from inhalation.
- A pre-flushing device is fitted to automatically purge the breathing circuit of N₂ with a burst of about 7 lit. Of O₂ when cylinder valve is opened.
- The apparatus is mounted on the back of the wearer.
- A dry ice cooler may be substituted for the saliva trap in the inhalation tube to cool the inhaled air temp to 15°C.
- Anhydrous NaOH (1.5Kg) is used as a CO₂ absorbent.

Air circulating system:

- Exhaled air containing CO₂ and excess O₂ flows through the exhalation tube, exhalation valve into CO₂absorption canister, freed from CO₂.
- The purified air flows into the breathing bag.
- Oninhalation, it is drawn from breathing bag into long demand valve housing where it gets enriched with O₂ and then flows through the inhalation tube into the wearer's wings.
- Excess pressure build up is prevented by the automatic relief valve.
- Additional O₂, if needed, is provided by the lung demand valve.

Dragger – BG-4

- It is a compressed – O₂, closed-circuit apparatus with a disposable soda-lime Co₂ – absorbent canister.
- It has been NIOSH-certified four times (1995) as both a 3 & 4 hr. apparatus and a standard and positive pressure apparatus.
- The four-hour positive pressure version of the BG-4 was tested using the same O₂ cylinder as that used for the BG-174A, which was the same recommended fill pressure of 3, 135 psi at room temperature.
- The three-hour version comes with a fiberglass-wrapped cylinder. The BG-4 has a constant flow of O₂ of at least 1-5 lit/min. **ATPD plus a volume-activated demand valve.**
- The relief valve is also volume activated and purged exhaled air before it enters the Co₂ absorbent canister.
- The face mask is the same as that used with the BG-174A.
- The wearer exhales into the face mask through the exhalation check valve and breathing hose, the CO₂-absorbent canister, and the spring-loaded breathing bag.
- Upon inhalation, the user draws air from the breathing bag through the heat absorbent canister, the inhalation hose, and check valve, and back to the face mask.
- The heat-absorbent canister works by routing inhalation air around a chamber containing a large cylinder of ice inserted before use, formed in a separate ice mold.



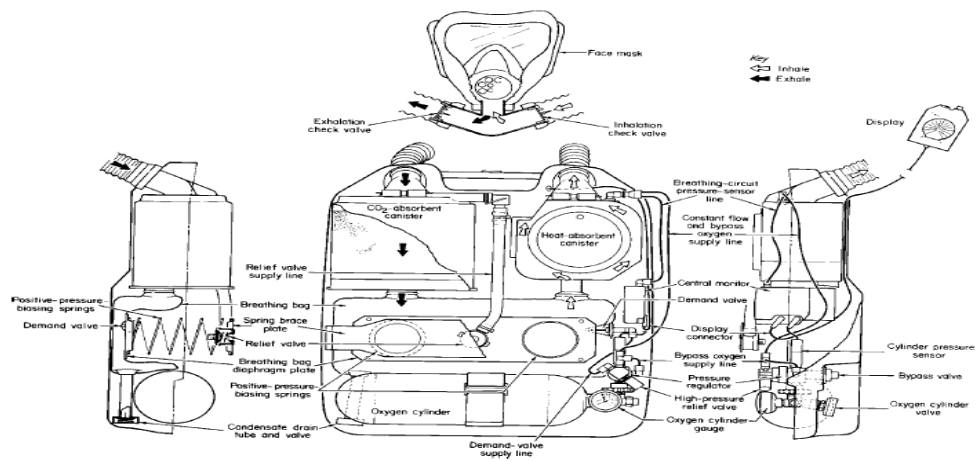


Figure: Photograph and schematic diagram of Dräger BG 4

Experiment-8

Aim: Study and sketch of reviving apparatus (Model: Pulmotor PT-60 and Carevent).

Apparatus Required :Pulmotor PT – 60 and Care Vent

Introduction: The apparatus was first constructed by Herishdragger in a year 1906 & in course of time was considerably modified and improved upon. It provides for modified automatic cycling of breathing by blowing as oxygen-air mixture into the lungs of the victim and removing the Co₂ charge exhausted air out of the lungs by suction with a simple adjustment, it can also be used as an inhalator.

Constructional features:It essentially consists of two 2.5 Ltrs. capacity steel cylinder containing oxygen at a pressure of 200kg/cm², a pressure reducing valve a distribution connection or manifold with three shut-off valves and a safety valve, a resuscitation or change over or cycling valve box, a face piece, an aspiratory injector, a secretion bottle, an oxygen economizer breathing bag, an inhalation face piece and three sizes of oral airways. The reducing valve reduces the high pressure oxygen to a constant operating pressure of 5kg/cm² of the three-shut off valve, one is connected to two aspirating injector, another to the oxygen inhalation mask, while the third id connected to cycling valve box. The cycling valve box is the soul of the apparatus.



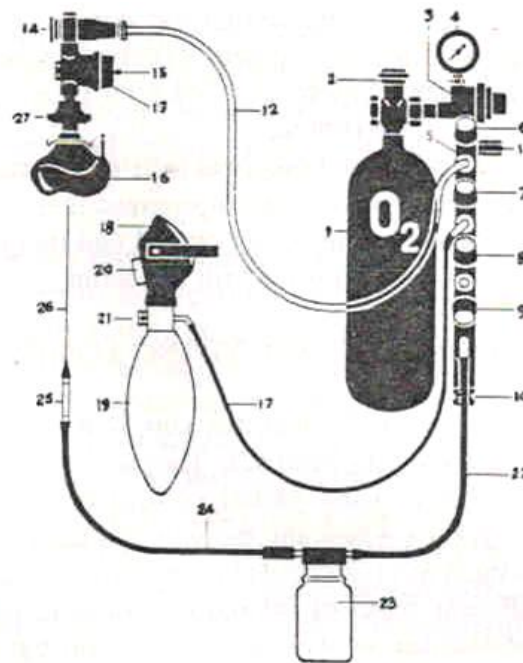


Fig. 7.28: Draeger Pulmotor 1 - oxygen cylinder; 2 - cylinder valve; 3 - reducing valve; 4 - pressure gauge; 5 - distributor manifold; 6 - pulmotor shut-off valve; 7 - inhalation shut-off valve; 8 - spare valve for connecting other equipment; 9 - suction valve; 10 - aspirator injector; 11 - safety valve; 12 - oxygen connecting hose; 13 - pulmotor switching unit; 14 - change-over valve for 50% or 100% oxygen; 15 - selector knob; 16 - facepiece; 17 - connecting inhalation tube; 18 - inhalation facepiece; 19 - economizer breathing bag; 20 - expiratory valve; 21 - air addition valve; 22 - connecting hose; 23 - secretion bottle; 24 - aspirator hose; 25 - secretion vision glass; 26 - aspirator catheter; 27 - respiratory valve (for cutting-off the negative phase), optional

Figure: Photograph and schematic diagram of Pulmotor PT60

Operating Process: During the inspiration phase, oxygen flows through an injector nozzle into the upper part of the valve box there by sucking atmospheric air through a spring loaded air regulating valve. The oxygen air mixture thus obtained flows through the floating needle valve into the lower part and there to the victim. As soon as the lungs are ventilation, a positive pressure builds up in the lower part of the valve box, and when it exceeds 1.50mm WG, it presses the control membrane upwards. In consequence, the hinge point of the two levers of the cycling mechanism is drawn above the cycling spring as result of which the floating needle valve and the air-regulating valves are closed and another valve on the junction side is opened. The injector now sucks the air out of the lungs of the patient and blows it into open through a valve situated on the top of the valve-box. As soon as negative pressure of 100mm WG is reached in the lungs, the control membrane is sucked downwards and the cycling mechanism brought to the position in the inspiration phase. A changing knob is provided to bring the cycling mechanism to expiration position at the beginning of resuscitation.

Points to note:

- During the operation of the Pulmotor, the wearer should always observe that the respiratory system remain constant.
- If, during resuscitation, the action of the heart can no more be observed, heart manage must be carried out during the expiration phase.
- Should serene occurs as hopper with CO poisoning the tongue should be immediately loosened after lifting the face-piece, and if necessary respiration interrupted?
- Pulmotor should not be used except as inheritor in case of poisoning by nitrous fumes.
- It should, however, be emphasized that the first few minutes after breathing has ceased are the most crucial and that manual artificial respiration must be started at once and continue until the Pulmotor becomes available.

CAREVENT



The O-Two CAREvent[®] CA has only one external control. The chemical agent and IDLH environments are considered to be so hazardous that the less the rescuer has to do to ventilate the patient, the quicker the patient can be resuscitated and the greater the chance for the patient's survival. With the O-Two CAREvent[®] CA, all that is required is to turn on the oxygen/air supply, apply the mask to the patient's face and secure the airway. The O-Two CAREvent[®] CA will provide automatic ventilation to the non-breathing patient at a rate and volume designed to provide adequate oxygenation while reducing the risk of gastric insufflation, in accordance with the current guidelines for resuscitation.

The micro-pneumatic circuitry of the O-Two CAREvent[®] CA requires no batteries or electrical supply to operate. Should the patient start spontaneous breathing, the demand breathing feature of the O-Two CAREvent[®] CA allows the patient to breathe at his or her own rate and volume. The patient's inspiratory effort (if adequate) will also cause the automatic circuit shut-off to stop the automatic cycling of the ventilator, allowing the patient to breathe at her or his own rate and volume. If the patient stops breathing again, the automatic cycling will restart with no action required by the rescuer.

A manual override button is provided on the O-Two CAREvent[®] CA to allow for the provision of two breaths followed by 30 chest compressions during CPR in accordance with the current resuscitation guidelines. In addition, it can also be used to supplement spontaneous ventilations if required. An additional feature of the O-Two CAREvent[®] CA is the two L/min mask purge, bleed flow. This reduces the risk of the patient drawing in toxic ambient air into the mask should a leak occur in the mask-to-face seal. By combining this simplicity of operation with technological sophistication, the O-Two CAREvent[®] CA provides trained individuals with a safe and effective means of maintaining artificial ventilation during

respiratory arrest, which may occur during confined space entry or a toxic chemical agent release.

Accessories available for the O-Two CAREvent® CA include the head harness system, which assists in maintaining a mask-to-face seal during extrication and transport and to provide hands-free operation.

Also available are the deluxe transport ventilator circuit with positive-end expiratory pressure (PEEP) port and the gurney clip, which allows the device to be attached to the handrail of the gurney or emergency stretcher during transport.

CAREvents® CA Specifications

Tidal Volume	500 ml
Breaths Per Minute	10
I:E Ratio	1 : 2
Automatic Flow Rate	15.0 L/Min
Manual Flow Rate	15.0 L/Min
Delay To Automatic Cycling Re-Start After Manual Button Depression	20 SECONDS
Positive Pressure Bleed Flow	2 L/Min
Auto Shut Off	5 – 8 Sec
Demand Breathing Flowrate	0 – 120L/Min
Inspiratory Resistance @ 60 L/Min	0 to – 6 cmH ₂ O
Input Pressure	45 – 87 PSI (3.1 – 6 Bar)
Maximum Airway Pressure	60 cmH ₂ O (58.8 mBar)
Operating Temperature	-18 to +50°C (0 to +122°F)
Storage Temperature	40° to +60°C (-40 to +140°F)
Input Connection	9/16" DISS
Patient Connection	15 / 22 mm
Patient Valve Dead Space	8 ml
Weight(Approx)	16 OZ / 0.45 Kg
Size	140 x 63 x 73 mm 5.5" x 2.5" x 2.9"
Cylinder Duration: @ 10 Bpm / 0.5 LitresVt	82 minutes

Experiment-9

Aim: Evaluation of performance of closed-circuit breathing apparatus using Artificial Lung Machine or Artificial Breathing simulator.

Apparatus Required : Artificial Lung Machine

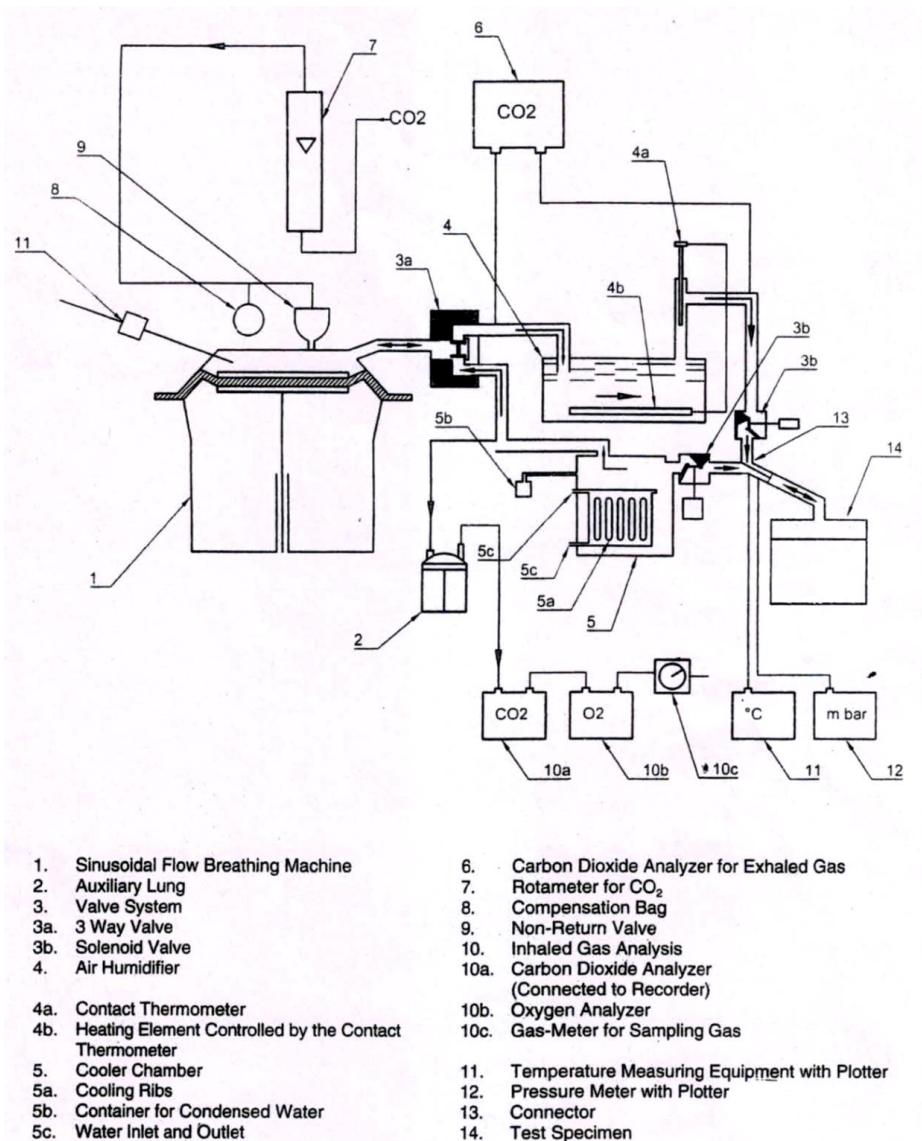
Theory:

Artificial lung machine is also called artificial breathing simulator. It simulates human breathing system. Machine sucks air from self-rescuer where it goes through two different sized piston cylinder arranged, which acts as artificial lung and main lung. Here CO₂ gas is added from an external cylinder to simulate the conditions of human respiratory system in which inhaled air consists of a higher value of carbon dioxide. This CO₂ consisting air is released into the self-rescuer in which the KO₂ granules react with the CO₂ present in the exhaled air and convert it into O₂. During this process following four parameters are studied.

- (i) Breathing resistance offered by apparatus.
- (ii) O₂ concentration of inhaled air.
- (iii) CO₂ concentration of inhaled air.
- (iv) Temperature of inhaled air.

The test is carried out according to EN 401 which is European Norms related to “Respiratory Protective devices-self-contained closed-circuit breathing apparatus for escape-Requirement, testing, marking”, confirming IS 15803.





Photograph and schematic diagram of Breathing Simulator

Test procedure:

Test is carried out in 3 batches, each batch consists of 5 samples. Few special features of test are as follows:-

- (i) Mass of complete apparatus including carrying case or container shall not exceed 5kg.
- (ii) Leak Thickness – The complete design checked by immersing in water.
- (iii) The opening pressure of relief valve shall not be less than 1 millibar at any condition.
- (iv) Breathing bag capacity or effective volume should be atleast 6 liter.
- (v) The rated duration of apparatus shall be at least 3 times longer than the rated capacity when tested with 10 l/min.

Important Notes:-

1. O₂ content in inhaled air shall not be below 21% by volume and a short term deviation to a level at not less than 17% by volume and for a period of not exceeding 2 min. at the beginning of the test is permissible.
2. CO₂ content throughout the rated duration of apparatus in inhaled air shall not exceed an average value of 1.5% by volume and shall not exceed individual value of 3% by volume.
3. In the apparatus with a rated duration of 15min., the temp. of inhaled gas shall not exceed 55°C when tested at 35 lit/min. during the rated duration.
4. Breathing resistance:-
 - (i) @35 lit/min. – For an apparatus of rated duration of up to and including 30min. the sum of the inhalation and exhalation resistance shall not exceed 16 millibar. Max. Individual breathing resistance shall not exceed 10 millibar.
 - (ii) @70 lit/min. – For apparatus of all rated duration the breathing resistance for inhalation as well as exhalation shall not exceed 20 millibar.